Original Article

Influence of Demographic Characteristics on Human Immunodeficiency Virus Prevalence among Intending Blood Donors in Orthopedic Hospitals in a Resource-limited Setting

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Abstract

Background: Human immunodeficiency virus (HIV) screening among blood donors is critical for ensuring blood safety, particularly in low-resource settings where transfusion demand is high. Orthopedic hospitals represent major consumers of blood products; however, data on HIV prevalence among intending blood donors in such facilities remained limited. Understanding the influence of demographic characteristics on HIV prevalence in this population is essential for risk assessment and targeted preventive strategies. **Objective:** This study aimed to determine the influence of demographic characteristics on the prevalence of HIV infection among intending blood donors at orthopedic hospitals in Anambra State, Nigeria. **Materials and Methods:** This was a cross-sectional study conducted among intending blood donors at selected public (n = 209) and private (n = 120) orthopedic hospitals in Anambra State, Nigeria. Demographic data were collected, and HIV serostatus was determined using standard screening protocols. Data were analyzed using Statistical Package for the Social Sciences version 25. Descriptive statistics were computed, and associations between demographic variables and HIV prevalence were assessed using the chi-square test. Statistical significance was set at a P value of <0.05. **Results:** The overall HIV prevalence varied across demographic groups. The highest prevalence was observed among donors aged 30–39 years (5.3%), female donors (7.3%), and single donors (4.4%). However, there were no statistically significant associations between HIV prevalence and the demographic variables examined, including age, sex, and marital status (P > 0.05). **Conclusion:** Although higher HIV prevalence was observed among donors aged 30–39 years, females, and single individuals, these differences were not statistically significant. These findings underscore the need for continued universal donor screening and reinforce the importance of demographic surveillance in maintaining blood safety in orthopedic healthcare settings.

Keywords: *Blood donation, demographic characteristics, HIV infection, orthopedic hospital*

Background

Human immunodeficiency virus (HIV) infection remains a global health concern, with 38.4

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million and 1.5 million persons living with and being newly infected with the virus, respectively. The majority of these HIV patients reside in low- and middle-income nations.[1]

From a global perspective, a study by Shrivastava *et al*.[2] noted that about 30 million blood products are transfused annually in these low- and middle-income countries annually. In addition, the prevalence of transfusion-transmitted infections was higher in rural areas due to illiteracy, poor healthcare resources, lack of proper diagnosis, awareness, and a limited number of studies in numerous geographical areas.[2]

Global distribution of coronavirus disease 2019 in 2020 posed a great burden on the health systems and invariably impacted blood donation, thereby contributing to the changes in HIV prevalence in many countries.[3]

Hence, blood transfusion has been impacted by the prevalence of HIV infection and, similarly, has led to approximately 1.6 million blood units donated by an estimated 92 million individuals worldwide being wasted annually due to the presence of infectious pathogens.[4] In order to prevent the transmission of these infectious organisms, the World Health Organization (WHO) recommended that collected blood units should be subjected to serological testing.[4]

In Pakistan, it was posited that every year, about 120 million units of blood are donated globally, making blood donors benefactors of humanity. However, badly screened blood can provide a high risk of transmitting blood-borne illnesses to recipients.[5]

Furthermore, it was observed that the transfer of blood-borne infections via blood transfusion is a major threat to humanity and a huge public health setback, particularly in developing nations.[6] In Nigeria and other sub-Saharan African countries, the situation is exacerbated by inadequate infrastructure, unqualified staff, and low financial resources.[6,7] In addition, several studies have noted a significant prevalence of HIV among ostensibly healthy individuals who wish to donate blood in Nigeria, Ghana, and the rest of Africa.[6]

In addition, the WHO report of 2020 indicated that more than two-thirds of the 37.6 million persons living with HIV worldwide reside in Africa, and transfusions of blood and blood products contributed to a high percentage of HIV infection.[7] Similarly, reports from Sub-Saharan Africa demonstrated that the overall median risk of contracting HIV via blood transfusion is 1 infection per 1000 units.[7]

Hence, the burden of transfusion–transmissible infections on individuals and the economy has led to the establishment of a global blood safety initiative by the WHO, which recommended that all blood donations should be screened for HIV, hepatitis B virus, hepatitis C virus, and syphilis.[8] However, HIV continues to pose a significant threat to transfusion medicine, particularly in Nigeria and other sub-Saharan African countries, where 10%–15% of HIV transmissions have been linked to blood transfusions.[8]

Nigeria, Uganda, and South Africa are reported to be responsible for about half of all new HIV infections in sub-Saharan Africa.[9] In addition, studies have indicated that transfusion–transmitted HIV infection remains high in sub-Saharan Africa among pregnant women due to hemorrhage-related anemia and children as a result of malaria-related anemia.[8,9]

The safety of blood transfusion has remained a serious public health issue in Nigeria due to inadequate national blood transfusion policies and services, infrastructures deficit, inadequate skilled staff, and poor funding. Therefore, robust screening of donors' blood, inquiring about their lifestyle and general health before blood donation, will reduce the risk of HIV transmission by blood transfusion.[10]

Hence, understanding how demographic characteristics influence the prevalence rate of HIV infection among intending blood donors at orthopedic healthcare facilities, who are major consumers of these blood products are vital for measuring the risk of transmission through the blood supply and adopting appropriate preventive measures.

Materials and Methods

Ethical approval

Ethical approval for the study was gotten from the ethics and review board of the National Open University of Nigeria (NOUN) with an approval code NOUN/20/10/2023/493. Study participation was voluntary, and written informed consent was obtained from all the participants before commencement. Anonymity and confidentiality of respondents were assured and maintained. The ethical principles of the Helsinki Declaration were upheld throughout the study.

Study design and study area

A cross-sectional study was conducted at two hospitals: public (Nnamdi Azikiwe University Teaching Hospital, Trauma/Orthopedic Center Oba, an annex of Nnamdi Azikiwe University Teaching Hospital, Nnewi) and private (Uzondu Specialist Orthopedic Hospital, Ojoo). Both institutions and communities are located

in Anambra State. Anambra State has a population of 4,177,828.[11] Both facilities offer a blood bank, a full laboratory, and admission services.

Sample population

The study was carried out among 329 healthy intending blood donors in public (209) and private (120) orthopedic hospitals in Anambra State, South East Nigeria.

Sample size and sampling technique

The sample size was calculated using the formula $n = Z^2 p (1 - p)/d^2$. A sample size of 245.08 donors was gotten and 10% attrition rate (25) was considered. However, a total of 329 (274 males and 55 females) blood donors were enrolled in the study. A convenience sampling technique was adopted for the selection of the study participants and the orthopedic healthcare facilities that were used for this study.

Eligibility criteria: Inclusion and exclusion criteria

Eligible participants were intending blood donors aged 18–65 years who presented to the blood banks of the selected orthopedic hospitals, both of which offer blood banking services. Participants were required to be apparently healthy at the time of presentation and to provide written informed consent following pre-donation counseling included in the study.

Participants were excluded if they reported a history of behaviors associated with increased risk of HIV infection, had evidence of acute illness or fever at the time of donation, or had a previously known HIV-positive status. Individuals who had received a blood transfusion within 3 months before enrollment were also excluded from the study. In addition, participants with incomplete demographic information or HIV screening results, as well as those who declined or were unable to provide informed consent after counseling, were excluded from the analysis.

Method of blood sample collection

Two research assistants were trained on data collection with a questionnaire, the collection of blood samples, proper handling of blood samples, adherence to ethical considerations, and adequate data collection.

Participants were asked to sit comfortably with their forearms extended. The specimen bottles are labeled with the patient's name and identification number. A tourniquet was tied about 5–10 cm above the cubital fossa, and the upper arm area of each participant was properly cleaned with cotton wool dipped in 70% alcohol solution, then wiped dry with a clean cotton wool. Five milliliters of venous blood were obtained from the antecubital vein. About 2.5 mL of the blood was introduced into a blood specimen bottle containing anticoagulant ethylene diamine tetraacetic acid to prevent the blood from clotting. The remaining 2.5 mL was transferred into plain tubes without anticoagulant according to the WHO standard method. The serum specimens were separated by centrifugation at 3000 rpm (revolutions per minute) for 5 min. The sera were stored at −20°C and used for serological analysis.

Examination of blood samples for HIV-1/HIV-2

An HIV test was done according to the national testing algorithm. Using the stored serum samples earlier collected from the participants, HIV screening was done with two kits as recommended by the WHO. Determine® HIV-1/2 (5 Abbott Laboratories, Abbott Park, IL, USA) was used for the test, whereas Uni-gold® HIV-1/2 (Trinity Biotech, Dublin, Ireland) was used for the confirmatory test. The HIV test strip Determine® was labeled with the client's identification number. The protective foil cover of the Determine® was pulled off. The test was carried out by applying 50 µL of the serum to the sample pad (marked by the arrow symbol) of the test unit and allowing it to lie flat on the bench for 20 min, after which the result was read and interpreted based on the presence or absence of two pink/red lines (control line and AB line). Negative samples are indicated by a reaction at the control window only, whereas positive samples are indicated by a double reaction at both test and control windows. Any reaction outside these is invalid. The rapid test kit has a sensitivity of 99.7% and specificity of 99.8%.

The sample that gave a positive result was further tested with the Uni-gold HIV test kit. The Uni-gold® HIV test device was removed from the package and labeled with the client's identification number. The stored serum sample was collected with a disposable pipette, and two drops (approximately 60 µL) of the specimen was added to the port in the Uni-gold device. The confirmatory result was read and recorded after waiting for 10 min. Negative samples are indicated by reaction at the control window only, whereas positive samples are indicated by double reaction at both test and control windows. Any reaction outside these is invalid. The Uni-gold rapid test kit has a sensitivity of 99.8% and specificity of 99.9%.

Data/statistical analysis

The collected data on demographic characteristics from the questionnaire were analyzed using descriptive statistics. Similarly, Statistical Package for the Social Sciences (SPSS) version 25 (SPSS Inc., Chicago, IL, USA) was used to calculate the chi-square to determine

the level of significance, which was set at 95% confidence interval ($P < 0.05$).

Reporting guidelines

This study adhered to the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guideline for observational studies to ensure completeness, transparency, and methodological rigor in reporting.

Results

Results in Table 1 show that a total of 329 intending blood donors, comprising 274 (83.3%) males and 55 (16.7%) females, were involved in the study. The age group of the intending blood donors was 20–29 years, 93 (28.2%), 30–39 years, 151 (45.5%), and above 40 years, 85 (27.1%), and the mean age is 35 years. There were more married 216 (65.7%) than single 113 (34.3%) intending blood donors.

Table 1: Demographic characteristics of the intending blood donors

Characteristics	Number of intending blood donors = 329	Percent (%)
Gender		
Male	274	83.3
Female	55	16.7
Age group (years)		
20–29	93	28.2
30–39	151	45.8
Above 40	88	27.1
Marital status		
Single	113	34.3
Married	216	65.7

Table 2: Overall prevalence of human immunodeficiency virus (HIV) among intending blood donors in orthopedic hospitals in Anambra State by age groups

Age groups	Number (%)	HIV-1 and HIV-2 positive (%)	HIV-1 and HIV-2 negative (%)
20–29 years	93 (28.2)	3 (3.4)	90 (96.7)
30–39 years	151 (45.8)	8 (5.3)	143 (94.7)
Above 40 years	85 (27.1)	3 (3.4)	82 (96.4)
Total	329	14 (12.1)	315

Table 2 shows that out of the 329 intending blood donors, 93 (28.2%), 151 (45.8%), and 89 (27.1%) were aged 20–29 years, 30–39 years, and above 40 years, respectively. Among the age group 30–39 years, 8 (5.3%) were positive for HIV, whereas 143 (94.7) were negative. Hence, the prevalence of HIV among intending blood donors aged 30–39 years was higher than that in other age categories.

The results in Table 3 show no significant difference ($\chi^2 = 0.708$; df = 2; $P = 0.997$) in the prevalence of HIV among intending blood donors in orthopedic hospitals in Anambra State by age group, and the P value is >0.05 level of significance ($P > 0.05$). This implies that intending blood donors of different age groups did not differ in their HIV status.

The result in Table 4 shows that out of a total of 329 intending blood donors involved in the study, 274 (83.3%) were males, whereas 55 (16.7%) were females. The prevalence of HIV shows that 10 (3.6%) males and 4 (7.3%) females were HIV positive. Consequently, there was a higher prevalence of HIV among female than male intending blood donors in Anambra State.

The results in Table 5 show that no significant difference ($\chi^2 = 1.476$; df = 1; $P = 0.191$) was found in the prevalence of HIV among intending blood donors in orthopedic hospitals in Anambra State by gender, and the P value was >0.05 level of significance ($P > 0.05$). This implies that male and female blood donors did not differ in their HIV status.

Among 113 single intending blood donors, 5 (4.4%) were HIV positive, whereas 108 (95.6%) were HIV negative. Thus, there was a higher prevalence of HIV

Table 3: Summary of chi-square test showing difference of human immunodeficiency virus (HIV) among intending blood donors in orthopedics hospitals in Anambra State by age groups

Variable	*N*	HIV-1 and HIV-2		χ^2	df	*P* value
		Positive O(E)	Negative O(E)			
Age group (years)						
20–29	88	3 (3.8)	85 (84.2)			
30–39	151	8 (6.5)	143 (144.5)	0.708	2	0.997
≥40	88	3 (3.8)	85 (84.2)			

O: observed outcome, E: expected outcome.

Significant at $P < 0.05$

among the single than the married intending blood donors in Anambra State [Table 6].

The results in Table 7 show that no significant difference (χ^2 =0 .012; df = 1; P = 0.559) was found in the prevalence of HIV among intending blood donors in orthopedic hospitals in Anambra State by marital status, and the P value was >0.05 level of significance. This implies that the single and married blood donors did not differ in their HIV status.

Discussion

The higher prevalence of HIV among intending blood donors aged 30–39 years is consistent with similar studies[9,10,12] with higher prevalence of 6.7%, 29.4%, and 6%, respectively

However, this study contrasts with other Nigerian studies[13,14] with a lower prevalence of 0% and 0% among donors of the same age bracket, respectively. In Bamenda, Cameroon, another study by the Nigerian Tourism Development Corporation[11] reported a lower prevalence of 2.5%, whereas at the National Blood Center, Yemen, a similar study by Aloy-Amadi *et al.*[12] recorded a lower prevalence of 0.3% among intending blood donors in the same age bracket.

In addition, at variance with these study findings are the reports of a study by Peliganga *et al.*[15] that recorded HIV prevalence of 4.3% among blood donors in the age group >50 years in Angola, whereas another study by Japhet[14] recorded a higher prevalence of 16.7% among blood donors in the age group >56 years in Ghana. In contrast, A study by Bhatti *et al.*[5] observed more donors within the age bracket of 18–27 years.

The higher prevalence of HIV in this study among intending blood donors in age group 30–39 years may be attributed to the fact that this age group

Table 4: Overall prevalence of human immunodeficiency virus (HIV) among intending blood donors in orthopedic hospitals in Anambra State by gender

Gender	Number (%)	HIV-1 and HIV-2 positive (%)	HIV-1 and HIV-2 negative (%)
Male	274 (83.3)	10 (3.6)	264 (96.4)
Female	55 (16.7)	4 (7.3)	51 (92.7)
Total	329	14 (11.0)	315 (95.7)

Table 5: Summary of chi-square test showing the prevalence of human immunodeficiency virus (HIV) among intending blood donors in orthopedic hospitals in Anambra State by gender

Variable	*N*	HIV-1 and HIV-2		χ^2	df	*P* value
		Positive O(E)	Negative O(E)			
Gender						
Male	274	10 (11.7)	264 (262.3)			
Female	55	4 (2.3)	51 (52.7)	1.476	1	0.191

O: observed outcome, E: expected outcome.

Significant at $P < 0.05$

Table 6: Overall prevalence of human immunodeficiency virus (HIV) among intending blood donors in orthopedic hospitals in Anambra State by marital status

Marital status	Number (%)	HIV-1 and HIV-2 positive	HIV-1 and HIV-2 negative
Married	216 (65.7)	9 (4.2)	207 (95.8)
Single	113 (34.3)	5 (4.4)	108 (95.6)
Total	329	14 (11.0)	315 (95.7)

Table 7: Summary of chi-square test showing difference in the prevalence of human immunodeficiency virus (HIV) among intending blood donors in orthopedic hospitals in Anambra State by marital status

Variables	*N*	HIV-1 and HIV-2		χ^2	df	*P* value
		Positive O(E)	Negative O(E)			
Marital status						
Single	113	5 (4.8)	108 (108.2)			
Married	216	9 (9.2)	207 (206.8)	0.012	1	0.559

O: observed outcome, E: expected outcome.

Significant at $P < 0.05$

represents the youths who are the most sexually active and are more likely to engage in high-risk behaviors like maintaining several sexual partners, engaging in casual sex and wild sex adventures, having unprotected sexual intercourse, tattooing, using intravenous drugs, and alcoholism that leave them susceptible to HIV infection. This observation is worrisome because the most vibrant and productive group of the population is the most affected.

The result of this study is consistent with several studies[6,15,16] in Sub-Saharan Africa that recorded a higher prevalence of 2.3% in females compared with males, 2.0% in Angola, recorded a higher prevalence of 12.1% among female donors in Ghana, and observed a higher prevalence of HIV in females than males in Kenya, respectively. In addition, other studies[17-19] found that among blood donors in Mali, the prevalence of HIV was greater in females 2.22% than males 2.15%, observed a higher HIV prevalence of 1.48% among females than males 1.20% at Dire Dawa Blood Bank in Eastern Ethiopia, reported a higher prevalence of 1.40% among female blood donors than males 0.79% in South Africa, respectively. Similarly, studies by Sharew *et al.*[20] and Omidkhoda *et al.*[21] recorded a higher HIV prevalence of 7.9% among the female blood donors than males 4.4% in North East Ethiopia, and a higher HIV prevalence among the female donors than the male donors at the Iranian blood transfusion center.

However, the findings in this study contrast with similar Nigerian studies[9,10,12,13] that recorded higher HIV prevalence among male donors. In addition, in other African countries, studies[11,22] recorded a higher HIV prevalence among the male blood donors 1.9% than females 1.6% in Bamenda, Cameroon, and a higher prevalence rate of 0.46% among males than 0.23% among females in Nairobi, Kenya. Similarly, in China, several studies[3,23] recorded a higher HIV prevalence among males 1.47% than among females 1.15% and the prevalence of HIV among male blood donors was higher than that among female blood donors, respectively.

The likely reason for more HIV prevalence among the female intending blood donors in this study may be as a result of the larger mucosal surface area in the female reproductive tract, which provides more opportunities for HIV to enter. Gender disparity and cultural norms that limit women's ability to negotiate safe sex practices, including condom use, may also be a contributory factor. Moreover, economic dependence and poverty can cause women to engage in transactional sex for financial support, which may increase their risk of contracting HIV infection.

Out of 329 intending blood donors involved in the study, 216 (65.7%) were married, whereas 113 (34.3%) were single. The result shows that HIV prevalence was higher among the single 4.4% than the married 4.2% intending blood donors, but the difference was not statistically significant, $P > 0.05$.

This study is in concordance with similar studies[9,10] in Nigeria, which recorded a higher HIV prevalence of 64.70% among single blood donors than 35.30% among the married donors in Port Harcourt, Rivers State, and reported a higher prevalence of 8.4% among single donors than married donors, 5.0% in Ilorin, Nigeria.

However, the findings are at variance with the reports[8,22] from other African countries, which recorded a higher prevalence of 2.55% among married than single donors, 1.6% in Bamenda, Cameroon, reported a higher prevalence among married 8.6% than single donors 1.8% at the Regional blood Transfusion Center in Kenya. Additionally, in Asia, another study by Boustani *et al.*[24] found a higher HIV prevalence among married donors than single donors at the blood donation facility in Ilam City, Iran.

The higher HIV prevalence recorded among the single than those with the married intending blood donors in this study may be attributed to the fact that the single blood donors are younger, indulge in more risky behaviors, more sexually active.

Strengths and limitations of the study

This study contributes to the existing literature by examining the influence of demographic characteristics on HIV prevalence among intending blood donors in orthopedic healthcare facilities, a population that has received limited attention in previous research despite being a major consumer of blood products. A key strength of the study is the inclusion of donors from both public and private orthopedic hospitals, providing a broader representation of the donor population. In addition, the study utilized a standardized and robust HIV screening protocol, thereby enhancing the reliability of the findings.

However, the study has some limitations. Its cross-sectional design limits the ability to establish causal relationships between demographic characteristics and HIV infection. The analysis was restricted to descriptive statistics and chi-square tests; more advanced analytical approaches, such as multivariable regression or correlation analyses, could have provided predictive insights and allowed adjustment for potential confounding factors. Furthermore, reliance on self-reported information for certain eligibility criteria may have introduced reporting bias.

Despite these limitations, the findings underscore the importance of sustained and rigorous donor screening practices to maintain blood safety. They also suggest the potential value of incorporating targeted HIV prevention and health education interventions within blood donation services, particularly in settings with higher observed prevalence.

Conclusion

The highest proportion of intending blood donors was observed among individuals aged 30–39 years, who also had the highest observed HIV prevalence. Although the donor population was predominantly male and married, HIV prevalence was marginally higher among female and single donors, respectively. These differences were not statistically significant and should therefore be interpreted as descriptive patterns rather than definitive demographic associations. The findings indicate that HIV remains an important consideration for blood transfusion safety and underscore the continued need for rigorous donor selection, screening, and preventive measures to minimize transfusion-transmitted infections, particularly in resource-limited settings.

Authors' contribution

BAO and IVO conceptualized and designed the study. BAO, IVO, and KOO were involved in data collection/acquisition and statistical analysis. All authors (BAO, IVO, EE, HNU, KOO, and AOO) were involved in writing and revising the manuscript for intellectual content. All authors read and approved the final manuscript and agreed to be accountable for all aspects of the work.

Ethical approval

The Ethical approval for this study was gotten on 20/12/2023 from the Institutional Review Board of the National Open University of Nigeria (NOUN) with an approval code NOUN/20/10/2023/493.

Informed consent

The study participants signed a written consent form. This informed consent form contained information about the names and affiliation of the investigator, description of the study in plain and simple language, duration of the study, and statement of confidentiality, including the right of the participant to withdraw from the study at any time.

Helsinki declaration

The present study was conducted by the authors in accordance with the ethical principles of the Helsinki Declaration.

Availability of research data

The corresponding author and all the authors are available and ready to supply the data upon any request.

Financial support and sponsorship

Nil.

Conflicts of interest

There are no conflicts of interest.

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